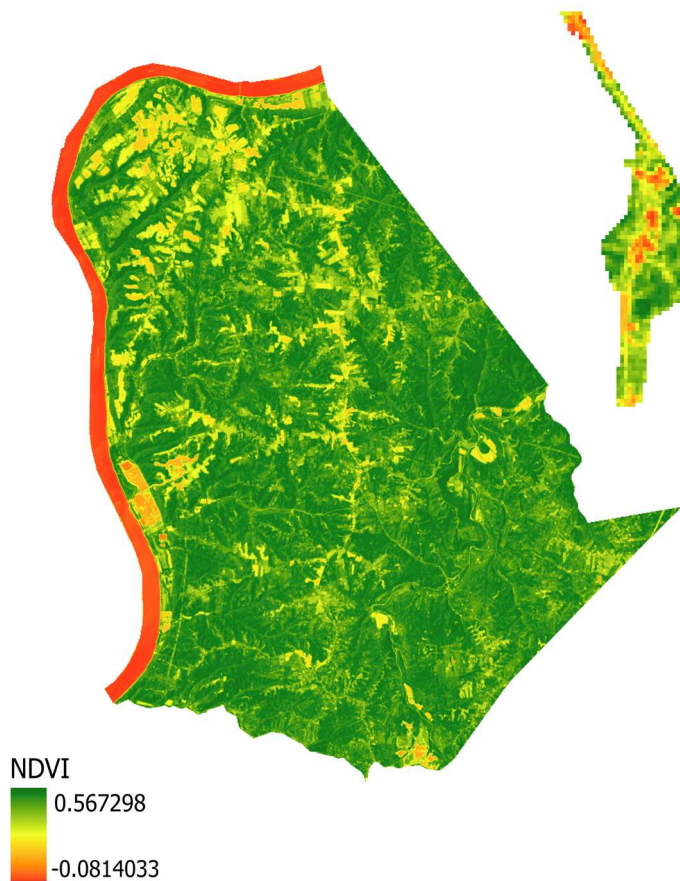


NDVI AND FOLIAGE GROUND COVER



The Trimble County Courthouse (right) serves as the heart of downtown Bedford and the county as a whole. However, it's lack of vegetation and natural shading makes it an uninviting place for visitors and members of the community. Planting more trees and creating a "shade garden" would help to make Bedford safer, as well

NDVI, or normalized difference vegetation index, is a way of measuring how much of an area's surface is covered by foliage using satellite imagery. The closer the NDVI value to 1, the greater the density of plant life.

While NDVI imagery captured from the Landsat 8 satellite (left) shows that Trimble County has a high level of vegetation, downtown Bedford is comparatively lacking. While Trimble's overall higher levels of vegetation can be attributed in part to a hilly landscape that discourages the removal of trees in areas too steep to build in, Bedford's lack of vegetation is still a critical shortcoming that needs to be addressed if the town hopes to see growth and future development as a community.

Extreme heat is the deadliest of all weather phenomena, and heat islands, which can be present even in areas like Bedford can exacerbate this issue without adequate shading. Studies show that ground cover that well-irrigated grasses, shrubs, and trees help to enhance the cooling capabilities of surfaces (Wu et al., 2021), as

